

Data Cleaning & Missing Value Handling

1.Introduction

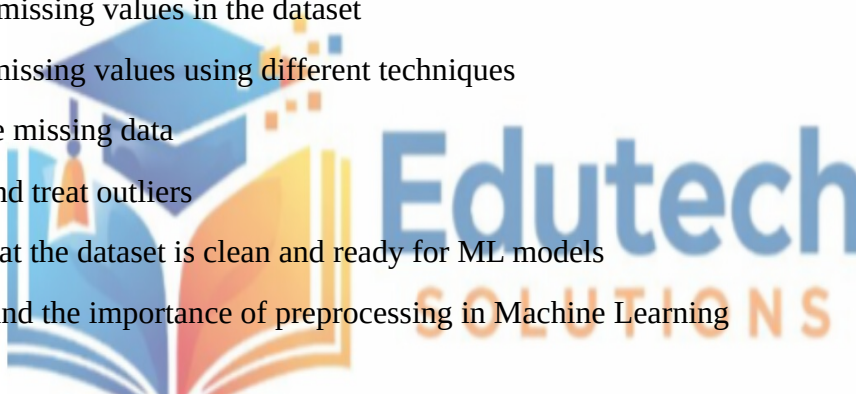
Data cleaning is one of the most important steps in the Machine Learning pipeline. Real-world datasets often contain missing values, duplicate records, incorrect data types, and outliers. If these problems are not handled properly, the performance of Machine Learning models decreases.

In this task, the objective is to clean a dataset by identifying missing values, handling them using suitable techniques, detecting outliers, and preparing the dataset for further analysis and model building.

The task was performed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.

2.Objectives

- Identify missing values in the dataset
- Handle missing values using different techniques
- Visualize missing data
- Detect and treat outliers
- Verify that the dataset is clean and ready for ML models
- Understand the importance of preprocessing in Machine Learning



3.Tools and Technologies Used:

Tool	Purpose
Python	Programming Language
Pandas	Data Manipulation
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Heatmaps & Graphs
Scikit-learn	Imputation Techniques

4.Dataset Used:

For this task, the **Titanic Dataset** (or Housing Prices Dataset) was used.

The dataset contains:

- Numerical features
- Categorical features
- Missing values
- Outliers

These characteristics make it suitable for practicing data cleaning techniques.

5.Steps Performed:

a. Importing Required Libraries

First, the required Python libraries were imported.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

b. Loading the Dataset

```
df = pd.read_csv("titanic.csv")
print(df.head())
```

The dataset was loaded successfully and the first few rows were displayed.

c. Checking Missing Values

```
print(df.isnull().sum())
```

This step helped identify columns containing missing values.

Observation

Some columns had null values due to incomplete data collection.

d. Visualizing Missing Values

```
sns.heatmap(df.isnull(), cbar=False, cmap='viridis')
plt.show()
```

Explanation

A heatmap was used to visualize missing values clearly.

This made it easier to understand which columns had more missing data.

e. Handling Missing Values

Different techniques were used based on the type of data.

Mean Imputation:

```
df['Age'].fillna(df['Age'].mean(), inplace=True)
```

Used for numerical data when the distribution is normal.

Median Imputation:

```
df['Fare'].fillna(df['Fare'].median(), inplace=True)
```

Used when data contains outliers.

Mode Imputation:

```
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

Used for categorical data.

f. Dropping Unnecessary Missing Data:

```
df.dropna(inplace=True)
```

Rows with too many missing values were removed.

Explanation

Dropping rows is useful when:

- The number of missing values is very small
- Missing data cannot be estimated properly

g. Detecting Outliers:

```
sns.boxplot(x=df['Fare'])  
plt.show()
```

Observation

Some extreme values were identified as outliers.

h. Handling Outliers

```
Q1 = df['Fare'].quantile(0.25)  
Q3 = df['Fare'].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5 * IQR  
upper = Q3 + 1.5 * IQR
```

```
df = df[(df['Fare'] >= lower) & (df['Fare'] <= upper)]
```

Explanation

The IQR (Interquartile Range) method was used to remove extreme outliers.

i. Verifying Cleaned Data

```
print(df.isnull().sum())
```

After preprocessing, the dataset was checked again to ensure there were no missing values.

Final Cleaned Dataset:

```
df.to_csv("cleaned_titanic.csv", index=False)
```

The cleaned dataset was saved as a CSV file.

6.Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load dataset
```

```
df = pd.read_csv("titanic.csv")
```

```
# Display dataset
```

```
print(df.head())
```

```
# Dataset information
```

```
print(df.info())
```

```
# Check missing values
```

```
print(df.isnull().sum())
```

```
# Visualize missing values
```

```
plt.figure(figsize=(10,6))
```

```
sns.heatmap(df.isnull(), cbar=False, cmap='viridis')
plt.title("Missing Values Heatmap")
plt.show()

# Fill missing numerical values
df['Age'].fillna(df['Age'].mean(), inplace=True)

# Fill missing categorical values
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)

# Drop unnecessary column
df.drop('Cabin', axis=1, inplace=True)

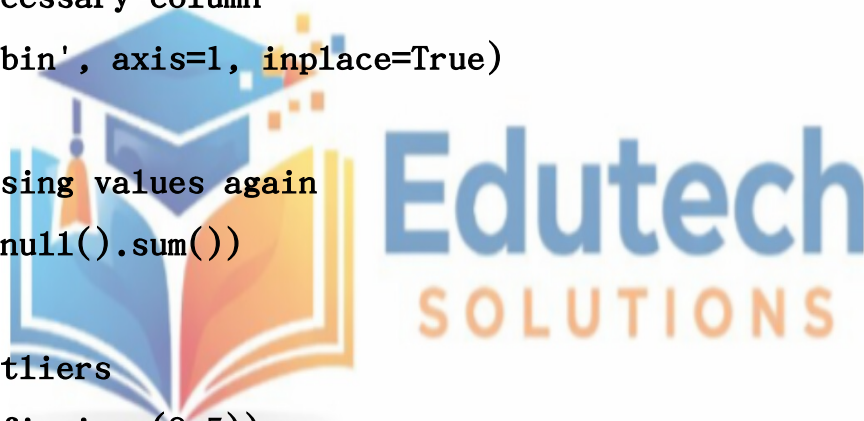
# Check missing values again
print(df.isnull().sum())

# Detect outliers
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Fare'])
plt.title("Outlier Detection")
plt.show()

# Remove outliers using IQR
Q1 = df['Fare'].quantile(0.25)
Q3 = df['Fare'].quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
```



```
upper_limit = Q3 + 1.5 * IQR

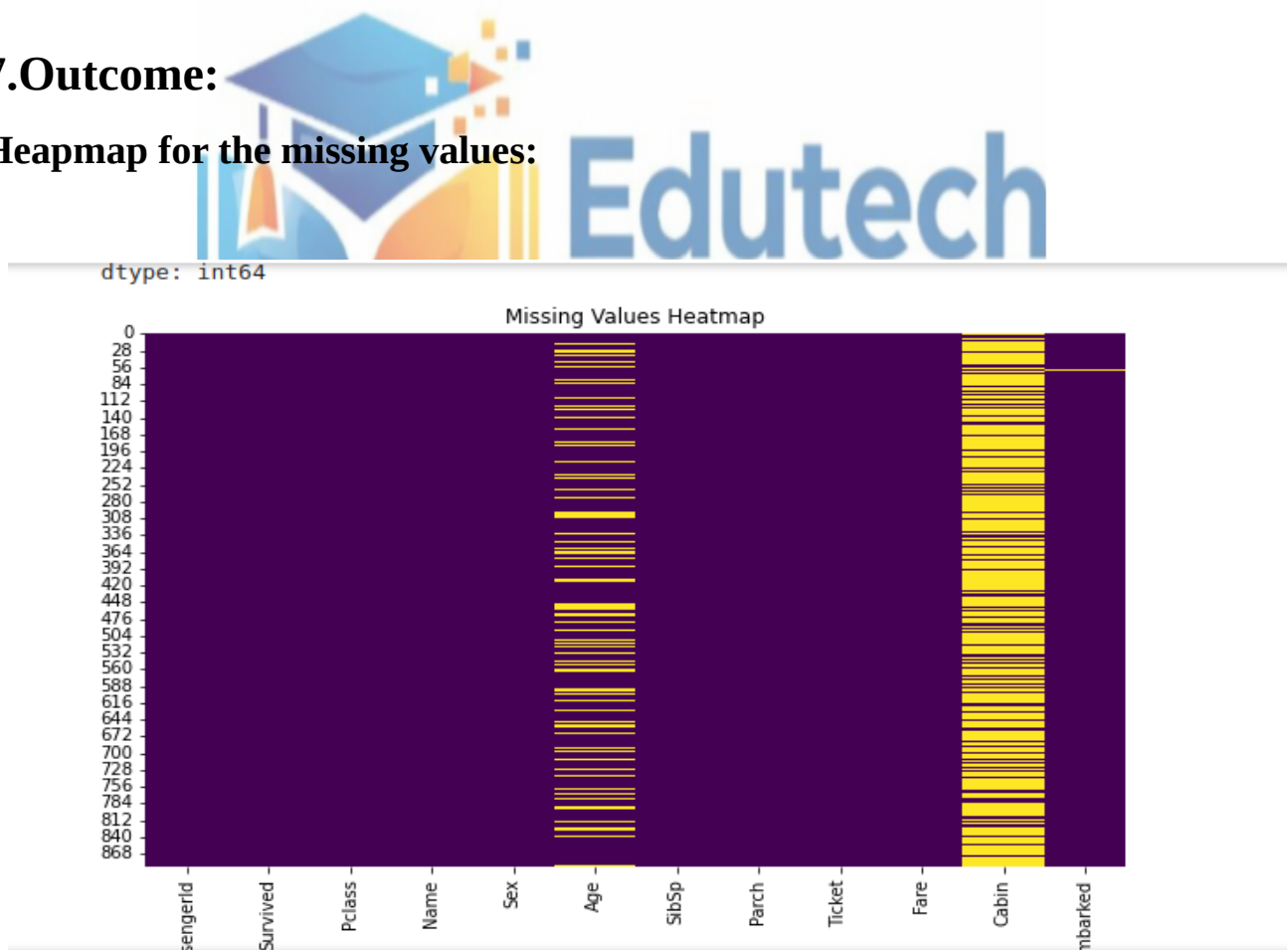
df = df[(df['Fare'] >= lower_limit) &
        (df['Fare'] <= upper_limit)]

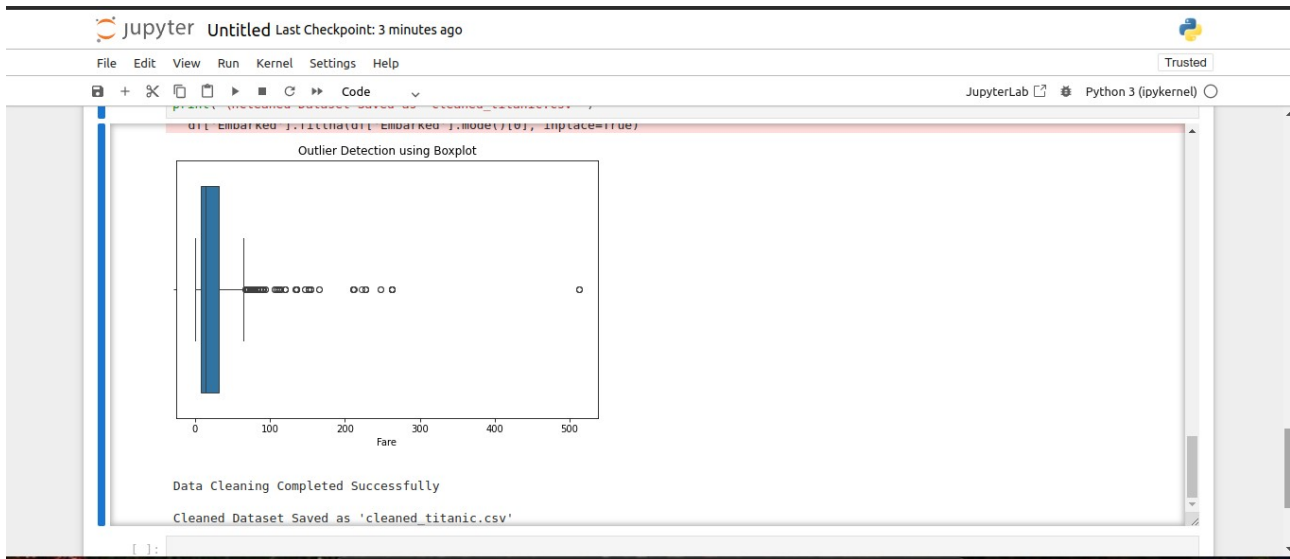
# Save cleaned dataset
df.to_csv("cleaned_titanic.csv", index=False)

print("Data Cleaning Completed Successfully")
```

7.Outcome:

Heatmap for the missing values:





Missing values after cleaning:

Missing Values After Cleaning:

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

8.Results:

- Missing values were identified and handled successfully
- Outliers were removed using the IQR method
- The dataset became cleaner and more suitable for Machine Learning
- Data quality improved significantly

9. Conclusion

Data cleaning is an essential step in data preprocessing. In this task, missing values were handled using mean, median, mode imputation, and row deletion methods. Outliers were also detected and treated effectively.

After cleaning, the dataset became more reliable for Machine Learning algorithms. This task helped in understanding the importance of preprocessing and quality assurance in real-world data science projects.

Interview Questions and Answers

1. What are the types of missing data?

MCAR (Missing Completely At Random)

Missing values occur randomly without any pattern.

MAR (Missing At Random)

Missing values depend on other observed variables.

MNAR (Missing Not At Random)

Missing values depend on the missing value itself.

2. When should we drop rows instead of imputing?

Rows can be dropped when:

- Missing values are very few
- Data loss is minimal
- Imputation may reduce accuracy

3. How do outliers affect imputation?

Outliers can distort mean values and lead to inaccurate imputation.

4. Difference between Mean and Median Imputation

Mean Imputation	Median Imputation
Uses average value	Uses middle value
Sensitive to outliers	Robust to outliers
Best for normal distribution	Best for skewed data

5. What is the role of data cleaning in ML?

Data cleaning:

- Improves data quality
- Reduces noise
- Increases model accuracy
- Helps better decision-making

Final Outcome

Successfully performed:

- Missing value handling
- Outlier detection
- Data preprocessing
- Dataset cleaning

The dataset is now ready for Machine Learning model development.

10.Task Summary

This task focused on Data Cleaning and Missing Value Handling using the Titanic dataset. Missing values were identified using Pandas functions and visualized through heatmaps. Different techniques such as Mean Imputation and Mode Imputation were applied to handle missing data effectively. Columns with excessive null values were removed to improve data quality.

Outliers were detected using boxplots and treated using the IQR method. After preprocessing, the dataset became clean, consistent, and suitable for Machine Learning applications. Finally, the cleaned dataset was saved as a new CSV file for future model development.

TASK 2

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Missing values handled successfully
Outliers detected and removed
Data quality improved
Clean dataset prepared for ML models

